Team 7

JAMN INC.

Janaki Bhosale
Noelle Gayer
Madeline Lowe
Abigail Thornton

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Part 1: Exploring the Problem Space

I. Problem Overview: Financial Awareness

a. Problem Description

In the United States, there is a significant educational deficit for personal finance [1]. Money drives our lives, whether we are willing to admit it or not. Financial status determines our ability to get an education, get a job, start and support a family, retire comfortably, and face the challenges of aging without fear of poverty. Despite the fact that finances are pivotal to the individual's overall health [2], Americans face a lack of education when it comes to their personal finances. Less than 50% of college students have taken a course to understand personal finance [3]. This statistic doesn't factor those not in college or Americans who never went college, indicating this statistic is likely generously high, meaning a great majority of Americans were not properly educated in financial independence and self-sufficiency [3]. This gap in financial education has societal impacts, as financially illiterate Americans are less likely to feel financially safe and successful.

b. Who is affected?

At a high level, this problem impacts United States citizens of all kinds. When discussing this issue within the team and members of various communities we are connected to, it became evident that college students are not the only demographic who are facing an education gap in financial awareness. Middle-aged and retired individuals also lack a robust education in personal finance, which can impact their ability to achieve professional and personal goals. In a study reviewed that examines financial literacy in the United States [4], statistics show that young adults and women are two of the most financially vulnerable groups, as well as citizens who did not graduate high school and those that are unemployed. In terms of race, White respondents had the highest correct response rate to the questions, indicating minorities are more impacted by this issue.

c. Initial Evidence Gathered

In evidence gathering, we read studies, made observations, and had conversations with peers to confirm that financial awareness is a problem space for many people. Firstly, from sitting in the studio presentations in class, it became evident that finance is a topic that impacts many college students. In describing three problem spaces, almost every group related back to technology that would support more financial awareness and agency. In addition, conversations team members had with their peers indicated that college students feel having access to personal financial education would be highly beneficial to them. Our team also read multiple papers about this domain. In one article, research shows that older generations are in search of more financial education and literacy in order to better plan for retirement [5]. Overall, researching papers helped the team gain a better understanding of the domain and those who are impacted by it [4].

d. Existing Solutions

The lack of financial education and literacy in America is being combated by a variety of solutions, both digital and otherwise. Digitally, many educational and financial institutions provide online resources that users can access and gain awareness of financial concepts. One example of this is Khan Academy, an education resource website, provides a free financial literacy course that is fully online [6]. This simple but extensive course is a great way for those who are not able to access/afford formal education to become more financially literate. Coursera is another online education resource that provides board access

to college and university sponsored courses on personal finance [7]. Fidelity, a financial services organization based in investing, has a learning page of their website where individuals can access information about investment, retirement plans, and other savings options [8]. The United States government also has online resources that can be utilized by the public to help gain greater financial knowledge [9].

II. Problem Overview: Budgeting

a. Problem Description

Budgeting is a key component in the broad domain of finances. Creating and maintaining budgets is essential for efficiently managing resources and when done improperly, can be detrimental for companies and individuals [10]. Budgeting can be a difficult task, especially if the individual lacks knowledge or experience with budgeting. Budgeting has a range of impact, from helping an individual save up for a dream experience or ensuring that lights and water always stay on. Therefore, it is important to be understood but unfortunately, there is a proportion of the population that do not know how to budget effectively [11].

b. Who is affected?

This problem impacts a significant number of adults (ages 18+) in the United States of America. The Pew Research Center conducted a study asking adult Americans to rate their confidence in ability to create and manage a budget which resulted in meaningful results about groups that are impacted by this issue. Low income Americans, minority Americans, and younger adults (age 18-29) are all groups that have much less confidence in budgeting than their peers [12]. College students are certainly impacted by this issue, as interviews with team members' peers revealed.

c. Initial Evidence Gathered

In evidence gathering, we read studies, made observations, and had conversations with peers to confirm that budgeting is a domain in which meaningful solutions will make a positive impact. Our team conducted interviews with friends and family members. Young adults (college-age) indicated that tools to make budgeting more simple positively impact their lifestyle by allowing them to be more confident in their financial decisions and have a better understanding of their financial position. Conversations with family members also revealed that better budgeting tools would have improvements in both small ways (clearer understanding of where money was being spent in a joint account) and larger ways (find ways to save up for international trips). Studies we read in this domain indicated that there is a need for more accessible, efficient budgeting [13].

d. Existing Solutions

The most simple and popular existing solution for budgeting is an Excel spreadsheet. Excel is utilized by individuals on a small scale and by companies on a larger scale. However, when discussing this solution with interviewees, it became evident that Excel has mixed reviews; some feel it is helpful and others find it to be clunky and unusable. There are a variety of other applications that are designed to make budgeting simple [14], which are reviewed generally positively. One popular solution for college students and young adults is You Need A Budget (YNAB), which helps make budgeting simple [15]. An interesting existing solution to the budgeting problem our team discovered in research is the concept of mental budgeting, which explores the human tendency to allocate their money according to specific categories, rather than

just viewing total income [16]. This research showed mental budgeting as an emerging but fruitful concept, which indicates there is still space in this domain for a solution.

III. Problem Overview: Financial Security

a. Problem Description

As discussed in Parts I and II, finances are a central part of the life of the individual. Thus, it follows that feelings of safety and security in finances are very important. Financial risks have increased in the digital era, ranging from cyber hacks into financial instructions from scams that trip people into giving their money or financial information into the wrong hands [17]. It is essential that Americans trust their online banking systems and feel confident in recognizing the difference between what is true and good and what is a potential fraud. Unfortunately, human users struggle in this domain, as they seek technology that is navigable and actively preventative of human errors that compromise their security [18].

b. Who is affected?

At a high level, almost everyone is affected by this problem. The majority of users of online systems do not have the expertise to effectively defend themselves against financial frauds or harmful cyber attacks, i.e. by creating strong passwords. Research highlights that older Americans (ages 60+) are the group most vulnerable to attack [19]. They lack the knowledge and experience to effectively recognize financial fraud and often pass sensitive information to the wrong hands. However, a study from the Pew Research Center shows that a large proportion of Americans remain vulnerable to attack, even though younger and more technologically capable individuals are better able to take action to mitigate long-term negative consequences [20]. In conclusion, the domain of financial security has a wide reach of individuals who can be positively impacted by technology designed to assist them in lowering their personal risk.

c. Initial Evidence Gathered

In evidence gathering, we read studies, made observations, and interviewed community members to highlight financial security as a societal issue that needs addressing. Conversations revealed that young people are willing to give their financial information online to a certain degree, but definitely take precautions when interacting with specific websites. However, older generations still feel unsafe when making purchases online, even though it has become socially necessary. An interesting observation from presentations in class during the studio session also indicate that many people feel like financial security is a domain that could use improvement, both in a broad sense through education and a smaller scope such as increased security measures that don't impact efficiency but do improve protections. The studies and papers we reviewed indicate that financial security is a concern for a significant number of people, and solutions seem vague and hard to use. Our team determined that there is definitely space in this domain to design secure and satisfactory solutions to this problem.

d. Existing Solutions

There are many banks and investing tools that have built in financial security as a part of their assurances and cyber security protection as an institution. A solution that's on the side of the user is LifeLock [21]. LifeLock does credit monitoring, activity alerts for 401K and investment activity on the users accounts, and can provide identity/social security alerts for the user. This can give a user the piece of mind that their personal information has an extra layer of protection and monitoring on top of the built in security of the institutions that they use.

Part 2: In-Depth Analysis of Chosen Problem Area

I. Problem Overview

a. Problem Description and Importance

In today's world, the potential for financial scams and fraud is extremely high risk. Regardless of age, everyone has the potential to be scammed, or for their privacy to be compromised. Across different platforms and methods, including social media, AI, emails, texts, websites, or deepfakes, there are a plethora of ways to be put in an unsafe position. When attackers successfully steal other people's information, it can result in financial losses that cause lost savings or a drained 401K, it can cause a strain on mental health as a result of depression and anxiety, and more. Scams and frauds are recognized and acknowledged by society, but awareness is not an effective mitigation strategy. In a high-stress environment or scenario where someone may not always be on alert, there is a chance the user does not have the instincts to catch on in the moment, meaning it is imperative to be more proactive than reactive in these instances.

b. Existing Solutions, Analog and Digital

DeepCover Game: https://deepcovergame.com/

- What the solution does well:
  - After downloading the app when you set up your profile there are options to use a "color blind mode", change the theme, and increase the volume of the effects or videos of the game. It has fun graphics and engaging puzzles for users and will be an interest to a younger user as they play an "agent."
- Where the solution falls short:
  - It is a mobile device app so there is a barrier of use immediately. While playing the game there is little to no financial security concepts being taught, it's just buzzwords like "wiretapping" being used to preface the situation before a user goes into a matching game.

ScamTeach.com: https://scamteach.com/

- What the solution does well:
  - It reviews various scam scenarios such as fake reviews on a product, phishing emails, and investment scams. It highlights the "red flags" of each scenario and at the end of each simulation asks the reader how many red flags they recognized (all, some, or none).
- Where the solution falls short:
  - It doesn't allow the user to point out what red flags they recognized until the very end so it might not be an accurate survey and should instead be placed while the user is being walked through each scenario and then being told if they have identified all the red flags or not.

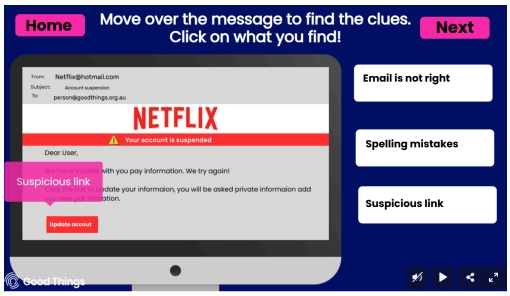
Tryscamme.app: https://www.tryscamme.app/

- What the solution does well:
  - It reviews both email and text scam messages and allows the user to try to identify what about the message is a scam. At the end it checks the user's guesses and gives the user a score. This solution also has the instructions on the side the entire time so the user can always reference back to them if needed to see the types of red flags and a reminder of the instructions.

- Where the solution falls short:
    - This could be a bug but the website wasn't saving the "role" that the user selected so there was an error and none of the phishing emails were shown. After many attempts there was success with the text message simulator. Some of the formatting with reading the messages was hard because a banner covered the top part of the simulator phone and zooming out was necessary to move from message to message. This can be a shortcoming for this application when older users are trying to engage and don't think to zoom out and are stuck in that one view.

Walkthrough of Existing System/Solution Example: GoodThings - Be a Scam Detective

- This Scam game gives users basic skills to spot scams through online social media messages and emails through phishing attempts.
- The user would be found by going to this link: Be a scam detective! Game | Good Things Foundation
- The user would write their detective name into a prompt and start the simulation. It has very simple controls with a back and next button at the top left and right of the screen respectively. In addition, it has simple instructions and appealing graphics for users to understand their choices in the game. This is great for an elderly user or someone who is younger and is learning about internet safety for the first time.
    - Example of instruction and scenario:



- NOTE: On the second social media scenario going through a suspicious instagram DM, the screen was unresponsive to my attempt to "click" on the clues it was pointing me towards. In order to move on I guessed on the quiz after the "investigation" on what was wrong with the message to move on. This is most likely a bug that the developers are aware of but it was something to note.
- Walking through this simulation as a user was interesting because there was only one social media scenario (Instagram Direct Message) and email scenario (Netflix payment inquiry) and it felt like there could have been more situations that a user could need to know about. Such as buzz words during scammer phone calls or other examples of social media/email phishing attempts. It almost seemed too short for the simplification of the game to get the point across. In addition, another feature that could be useful to acknowledge at the end of the scenario is how to deal with the message/email (blocking and reporting the account that directly messaged the user/reporting the email as phishing or deleting the emails safely).
- Overall, the game is an interactive way to teach users about clues when messages and emails can be possible scammer attempts to get information. Its simple controls allow the users to be directed

easily into each scenario where the clues and suspicious behavior are clear and users can easily read and use the system.

II. Stakeholders and Users

a. Stakeholders, User Groups and Their Characteristics

Due to the nature and scope of the problem domain, the primary stakeholders and users are anyone who uses the internet. However, individuals of different ages and with varying interests are at risk of becoming victims through different means. Outlined below are some different user groups of the internet and how they can be impacted by common scams:

Immigrants and Non-Native English Individuals: Members of ethnic communities within the United States, are unable to speak English, and do not have much knowledge on government agencies and businesses. The group will typically be uneducated and unaware of what scams or financial crimes may look like. By posting posters in community dense areas or on social media sites, they may be unable to recognize they are going to be scammed. [22]

Racial Minorities: Racial minorities make up a large population of the United States citizenship and these groups are generally equally likely to be targeted for fraud. However, some of these groups have less access to higher education and personal finance resources, leaving them more vulnerable to these targeting attacks. According to data from the Pew Research Center, American adults that identify as racial minorities are more likely than white adults to experience multiple forms of online financial fraud [20].

Social Media Users: People from all generations use social media; younger generations spend more time on applications like TikTok and Snapchat, while older generations utilize applications such as Facebook. Social media is a breeding ground for all types of fraud, not just financial fraud. Data shows, however, that in 25% cases where individuals lost money in fraudulent scams, the interaction happened on social media [23]. Social media users are characterized by low trust in the news content they see, but this distrust does not always extend to the other seemingly normal users they interact with [24]. One example of this is users having productive conversations on Reddit, even if they cannot physically interact with the other users [27].

Older Adults: Seniors, defined as age 65 and older, are more hesitantly adapting to the technology boom that society has experienced in the past twenty years. However, about 60% of senior United States citizens are online [25]. An interesting discrepancy exists within this group however, because while 77% of senior citizens have cell phones, only 47% have high-speed internet at home [25]. While their usage and habits may look different, this generation of Americans is online and is therefore vulnerable to financial scams.

Neurodivergent Users: Neurodivergence is defined as users with developmental conditions such as autism or dyslexia. For these users, technology is more usable when it is predictable and low-stimulating [26]. Unfortunately, these conditions are not common in application design today, which means that neurodivergent users are another group that can become confused or overwhelmed when interacting with mainstream technology and therefore more vulnerable to financial scams and fraud through technology.

This list is not exhaustive; as aforementioned, it is difficult to find user groups who do not interact with technology in some form. The groups listed above are meant to represent an intersection of the wide range of user groups who interact with technology and the groups who may be facing the highest financial risk online.

III. Tasks and Context

a. Key Tasks Users Perform

Users have various tasks they want to accomplish in the domain of financial security:

1. Utilize online banking services securely
2. Manage passwords and authentication methods
3. Make informed decisions that protect financial safety
4. Gain awareness of common fraud schemes and techniques
5. Recognize and report fraudulent activity
6. Minimize damage from scammers or cyber attacks
7. Understand measures that inherently exist in online transactions to ensure security

b. Characteristics of Tasks

The tasks described above are defined by a variety of characteristics:

1. Time-sensitive: recognizing and defending against fraud or cyber attacks must happen quickly in order to successfully mitigate the extent of damage.
2. High-pressure: protecting financial information is very important to users, so these tasks have a heightened sense of pressure to be successful.
3. Limited room for error: making mistakes on user tasks in this domain can be highly damaging to individuals and organizations.
4. Critical thinking: successfully determining the difference between legitimate and fraudulent schemes is important and takes critical thinking skills like deduction and reasoning.
5. Ease of use: all of the tasks above should be easy for users with limited prior experience and knowledge to confidently complete.

c. Task Environment

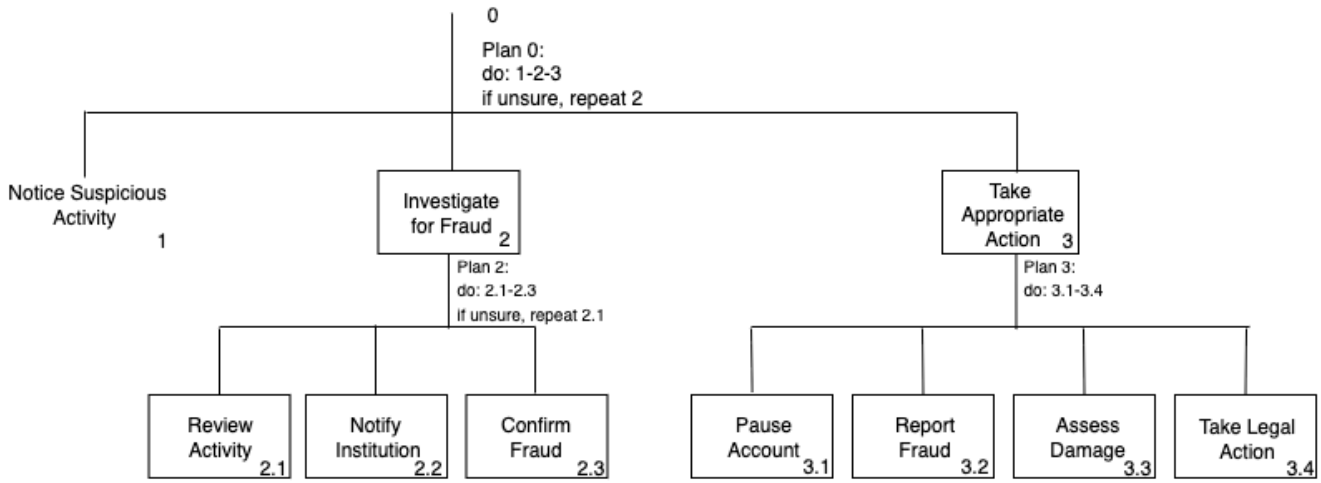
Physical: In the physical task environment, users typically interact with financial tasks on personal devices such as phones and laptops as well as with ATMs in public and private. Financial safety risks can increase based on the user's privacy and other's ability to see their screen, as well as the user's focus on the task at hand being affected by their surroundings.

Social: In the social task environment, important financial tasks are usually done alone, but can sometimes become subject to external pressures and be led to make risky moves with their finances. These external pressures could be from spouses or family or general societal pressures, but can also come from scammers who lead them to make hasty decisions with increased urgency.

Organizational: In the organizational task environment, user tasks fall within the scope of external organizations, such as banks or shopping centers, whether they are online or physical spaces. Users are constrained by any regulations and deadlines set by organizations and are also impacted by placing their financial security in the hands of any entity that they exchange money with.

d. Structured Task Analysis

Figure 1: Hierarchical Task Analysis (HTA) of Task 5



IV. Research Methods and Evidence

a. Data Gathering Methods

To gather information relevant to the chosen domain, two primary research methods were used. All team members researched scholarly articles to explore the problem, identify relevant information, and examine existing solutions and prominent gaps or areas for growth and improvement. Second, after deciding to focus on the financial security aspect, team members conducted interviews with multiple individuals who were considered to be relevant stakeholders, including: college peers and family members. An example of a group of stakeholders interviewed was college students who had little to no background in financial awareness or practices and information gathered through standard questions with followed up reasoning for why they had their opinions and what their experiences stemmed from.

b. Evidence Explored

In exploring evidence for this milestone, our team conducted interviews with members of our community (peers and older family members) and utilized scholarly research made available by the ACM Digital Library. Citations of this research are appended to this document in a Reference page.

V. Synthesis and Design Implications

a. Lessons Learned

The user base is expansive which allows for many facets of information to be lost to certain user groups if they aren't taken into account. In addition, it means that what is common sense for a younger user base technology wise is difficult for an older user base and vice versa when dealing with general knowledge of financial intricacies (Stock protections, informed decisions surrounding financial security). In addition, the tasks that are important to each user group vary by age and knowledge as well, middle aged users

want to have an awareness of fraud schemes that involve them personally being a key to the exploitation of their account. Meanwhile, a younger user could find importance in the bank's security system and projections of the programs it provides from cyber attacks. Finally there is such an expansive amount of resources and systems to help financial security from banks, cyber security companies, and personal good practices by a user that can provide a safe feeling surrounding their financial well being. This can be overwhelming due to the sheer amount of information and wanting to make sure everything is correct and lined up with the users accounts and information (causes of stress and anxiety if unclear). Overall this space has a wide range of users, tasks, and systems that are intersecting which allows for a variety of solutions and means that there is space for our group to create a bridge in an area that is lacking resources.

b. Implications for Future Design Decisions

These findings have lended the team to look into a niche of financial security. A possible design decision could be geared towards multiple user groups (young, old, uneducated in financial security practices) and focus on a specific niche like a task (being able to recognize fraud activity and awareness of common fraud schemes and techniques). To accommodate these user groups our design would have to be aware of the lack of technological savvyness and accessibility of the older generation and the interest and attention of the younger generation to keep them engaged to learn.

c. Constraints and Opportunities

A desktop web application would be the most appropriate so that it can be accessible to older generations without having to "download another app" and can be viewed with only an internet connection which is sometimes what a user group only has access to. As stated above, the UI should be simple enough to provide direction for older folks so they don't get overwhelmed with complicated processes. In addition, due to the short attention spans of a younger generation the UI should be engaging and with limited fancy jargon of financial security to not lose the reader's ability to understand the material and turn them away from the application. Finally, the application should be accommodating the knowledge that a user has of financial security going into using the application, ranging anywhere from knowing nothing to having a good idea but wanting a refresher.

VI. Peer Feedback and Reflection

a. Key Themes in Feedback

Upon reviewing the peer feedback we received there were a few clear patterns that presented themselves. The main theme that we noticed was a concern about broad scope, mainly in reference to the safety problem space (Idea 2). A lot of our peers believed that it had too much breadth and could be much too complex for a semester-long project unless it was narrowed down even more than the existing solution suggestions we presented. The concern for specificity was another theme that emerged in the feedback. Many peers were worried that even the more specific problem space suggestions were still too broad for every proposed problem area, so that is something that going forward we are paying a lot of attention to. The last major theme we noticed was that our peers were worried about the oversaturation within our first proposed problem space of finance (Idea 1). They were unsure if there would be room for us to create in a domain filled with different applications.

b. Identified Strengths and Weaknesses for Each Problem

Our first proposed problem space, finance (Idea 1), was seen as being the best and most feasible choice for a semester project, which is a big strength. Some of our peers were concerned about it being an oversaturated space, but ultimately they thought there was room for improvement, and that the fact that there were many existing financial applications could be a great way to start on our project, so the weaknesses listed are actually strengths as well. The second proposed problem domain, safety (Idea 2), was the one that our peers were the most excited about because they thought that its strengths were its relevance and potential to be impactful. However, our peers also found a lot of weaknesses as well, such as the fact that physical safety is hard to work with in a digital space. There were also many concerns about how broad the scope could be and working within a semester and how we would be able to test safety in the future. There was a lot less feedback on our third idea, mindfulness (Idea 3), which is a weakness in itself as our peers were not particularly interested in it and did not think it would be impactful. The ones who did respond had worried that such a broad scope could be a weakness and thought it would be hard to benefit mental health in a meaningful way in this project, as it would require a lot of research. There was very little feedback about the mindfulness space, likely because we stated a particular interest in finance and safety, and the fact that a lot of technology is coming out within that problem space.

c. Peer Feedback's Influence on Choice

The feedback we received from our peers really helped solidify our choice to explore the finance problem space. Firstly, it was the only idea that did not receive a single "Disagree" on whether or not our peers thought the problem space supported "at least 3 distinct and viable solution directions." A majority of our peers saw the finance problem space as the most feasible as a successful semester project, and even though there were concerns about oversaturation, some thought it would be helpful to have existing technology to help us explore some gaps within the space and inspire us to create something unique. There was little excitement about the mindfulness space, which helped us drop it as an option. Lastly, even though multiple peers were excited about safety, it was surrounded by a lot of uncertainty on how viable it would be as a semester project. Using this feedback, it was easy to support our ultimate decision to move forward in the finance problem domain.

d. Point of Disagreement or Tension

There was not much conflict in the feedback, but a few small points that we decided not to follow. Firstly, we could not do all three domains, so we did not follow with suggestions for safety and mindfulness. There were multiple concerns about oversaturation within the space as well, and we chose to overlook that concern and instead use it to our advantage, as we can find holes within the existing technology and create something new, while also referencing them as a starting point in our implementation research. We also believed that there were perhaps altogether too many concerns about broad scope within domains given the nature and purpose of the presentation, but tried to take them into consideration a little bit in regard to feasibility.

e. Future Work Implications

Ultimately the feedback we received will shape our approach moving forward by highlighting concerns to take into account as we go deeper into the project. Firstly, we will make sure our problem space is specific and defined, so that the responsibility and users will be clear and easier to test. Next, we will make sure to

the best of our ability that our project is creative and different in a space that is seen as oversaturated. Lastly, we took into account that some of our classmates were excited about safety and mindfulness, so we are working on finding specific spaces for those within the problem domain of finance in order to create something helpful and impactful.

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